# Camunda DISP Testing

# Test Strategy

For the testing, we created test cases based on the user forms and service tasks of the process. Each test case has an expected result that the system should ideally do when run, as well as the actual result of the test case, which didn't always match up with the expected result. The status of each test case in the "Goal Achieved" column is based on the following key:

- Yes: The actual result and expected result match almost perfectly
- Partially: Some aspects of the actual result match with the expected result, but there are key areas where the actual result does not meet what was expected
- No: The actual result does not meet the expectations of the test case.

# Test Table

| Test Case | Expected Result | Result of the Test | Goal Achieved |
|---|---|---|---|
| Test 1: Select product | The form takes the product category, product name, quantity and if it's for hire or sale | The form took the input fields and submitted accordingly | Yes |
| Test 2: Use tool confirmation | The form takes the tool name, expected return date and time, tool condition | The form took the input fields and submitted accordingly | Yes |

| | | | |
|---|---|---|---|
| Test 3: Calculate the price | The price is given by the database and it multiplied by quantity and it returns a total | There is no database, but a default price is multiplied by quantity | Partially |
| Test 4:<br>Check the availability | The process checks a database for the availability of an item | There is no database | No |
| Test 5: Delivery, click and collect or direct purchase | User is required to select an option for collecting their purchase, and only needs to give an address for delivery | User can select an option, but has to give an address no matter what, and collection time field can be filled with non time formats | Partially |
| Test 6: select payment method | The form takes payment method selection(card,finance and cash but credit and card number is only required if card is chosen, trade card number is only required if finance is chosen) | Credit card number is required no matter what choice of payment is chosen | Partially |
| Test 7: process payment | Payment is successful | Payment is successful | Yes |
| Test 8: Validate if user is a fintrust member | User is rejected if they do not give a trade card number or put 0 as the number | User is sometimes rejected no matter what, sometimes accepted | No |
| Test 9: Fintrust pays for the customer | Payment is done by Fintrust if user is a valid trade card member | Payment is done by Fintrust if user is a valid trade card member | Yes |
| Test 10: Customer repayment | Simulate repayment by calculating the required monthly payment | Repayment is calculated by dividing total price by 12 | Yes |
| Test 11: Stock level update | Java code gets the product name and quantity and updates stock in the database | Java code gets product name and quantity, but there is no | Partially |

| | | | |
|---|---|---|---|
| | | database. Updated stock level is simulated instead | |
| Test 12: Create shipment | Shipment ID is created, and shipment info is logged | Shipment ID is created, and shipment info is logged. Only happens if the user has selected delivery | Yes |
| Test 13: Send delivery notification | Simulated notification by logging customer info, shipment ID and the product | Simulated notification by logging customer info, shipment ID and the product, but only the customer ID, shipment ID and product name is logged. No delivery address, collection time or product category is logged | Partially |
| Test 14: Use tool form | All fields are required. Agreeing to tools and conditions, date and time, and tool condition. Product name is prefilled and cannot be changed | All fields are required, product name is prefilled but can be changed | Partially |
| Test 15: Categorise tool: | User inputs whether the tool needs routine maintenance, repairs or is completely out of service | User inputs whether the tool needs routine maintenance, repairs or is completely out of service | Yes |
| Test 16: Decommission tool | Log info of the tool along with product name and customer ID | Log info of the tool along with product name and customer ID | Yes |
| Test 17: Generate report | Generate reports including product information, and customer ID, simulated by logging them | Product name and customer ID are logged, but not quantity or product category | Partially |

# Conclusion

For the most part, the entire business process is accurately simulated, however there are some areas that can be improved.

The biggest improvement that could be made would be the addition of a database. This could store product information and some trade card numbers that could be used for validation. The “Check Availability” service task could look for the input product name in the database, and return whether it is available or not depending on whether it is in the database, and if so, whether the quantity requested by the customer is more than or equal to the quantity in the database. Stock levels can then update these quantities, rather than simulating it. Additionally, the FinTrust Pool can check the database for valid trade card numbers, then validate whether the user is a valid member based on that.

Another improvement that could be made would be extra additions to the forms, such as some fields only being required if specific options are chosen. For example, if the customer selects click and collect or direct purchase as the collection option, they should not be forced to input their address. However, if they do, the address field should be validated by checking for the correct format, rather than just validating based on input length. Additionally, the collection time input should be a date and time format, rather than a text field. This way, only a date and time could be input, rather than the process accepting any input.